Module 13: How Populations Evolve - Learning Questions

1. What is an allele frequency?
2. How is microevolution measured?
3. Which mechanisms can change allele frequencies?
4. How does mutation differ from gene flow?
5. Why is genetic drift stronger in small populations?
6. How do bottleneck and founder effects differ?
7. Why is natural selection not random?
8. How do directional, stabilizing, and disruptive selection differ?
9. What does Hardy-Weinberg equilibrium represent?
10. How can a null model help detect evolution?
11. A diploid population contains 20 AA, 40 Aa, and 40 aa individuals. Calculate p and q, showing allele counts. Under Hardy-Weinberg assumptions, what genotype frequencies would you expect?
12. Can nonrandom mating change genotype proportions while leaving allele frequencies unchanged? Explain why a Hardy-Weinberg mismatch alone does not identify the mechanism.